RESEARCH REPORT · ITERATION 1 · 26 OCT 2026

CV EL research iteration 1

Results of the first round of experiments on the CV EL Alzheimer's project. We repaired the original model, measured how much the original dataset leaks, and validated the statistics for the planned genetics-vs-MRI paper on synthetic cohorts. The leakage is now measured, the original SNN transfer is confirmed broken, and one promising idea failed. No clearly novel contribution has emerged yet. The main blocker is still access to ADNI data.

Scope and data rules

- **Real data used:** the original Kaggle dataset (44,000 2D slices, Apache-2.0) and the trained model `alzheimer_cnn.h5`.
- **Synthetic data:** simulated cohorts and volumes for E03, E05 and E07. These check that code and statistical tests behave correctly. **They are not clinical evidence.**
- **No patient-level MRI or genetic data** have been analysed yet. ADNI access hasn't been requested.
- The original notebook, model and paper are unchanged, verified against SHA-256 hashes. All 36 tests pass.

Summary of experiments

ID	QUESTION	ANSWER	DATA
E01	Can the trained model be moved into PyTorch correctly?	**Yes. 100% prediction agreement; probabilities within 0.0018**	Real model, 64 slices
E01	Was the notebook's SNN weight transfer correct?	**No. 17% agreement (chance is 25%); no biases or classifier weights copied**	Real model
E02c	Does the image-level split leak?	**≥ 67.3% of validation images have a near-copy in training; 1.5% are byte-identical**	Real (Kaggle)
E08	How much accuracy does that leakage buy?	**At least 5.6 points (79.3% vs 73.6%)**	Real (Kaggle)
E03	Is the genetics test valid with a learned MRI score?	Only with out-of-fold scores; it can't separate mediated from direct genetic effects	Synthetic

ID	QUESTION	ANSWER	DATA
E04	Do the calibration metrics give known answers?	Yes	Synthetic
E05	How should saliency-anatomy overlap be tested?	Shuffling voxels: 34.5% false positives. Shifting the mask: 5.0%	Synthetic
E07, E07b	Can double machine learning isolate a direct genetic effect?	Not at realistic sample sizes (44% false positives at n = 400)	Synthetic
E06	What does the M5 deliver for 3D training?	GPU about 2× CPU (3.5 volumes/s); fp16 gives no gain	Synthetic tensors

1. The original model and SNN conversion (E01)

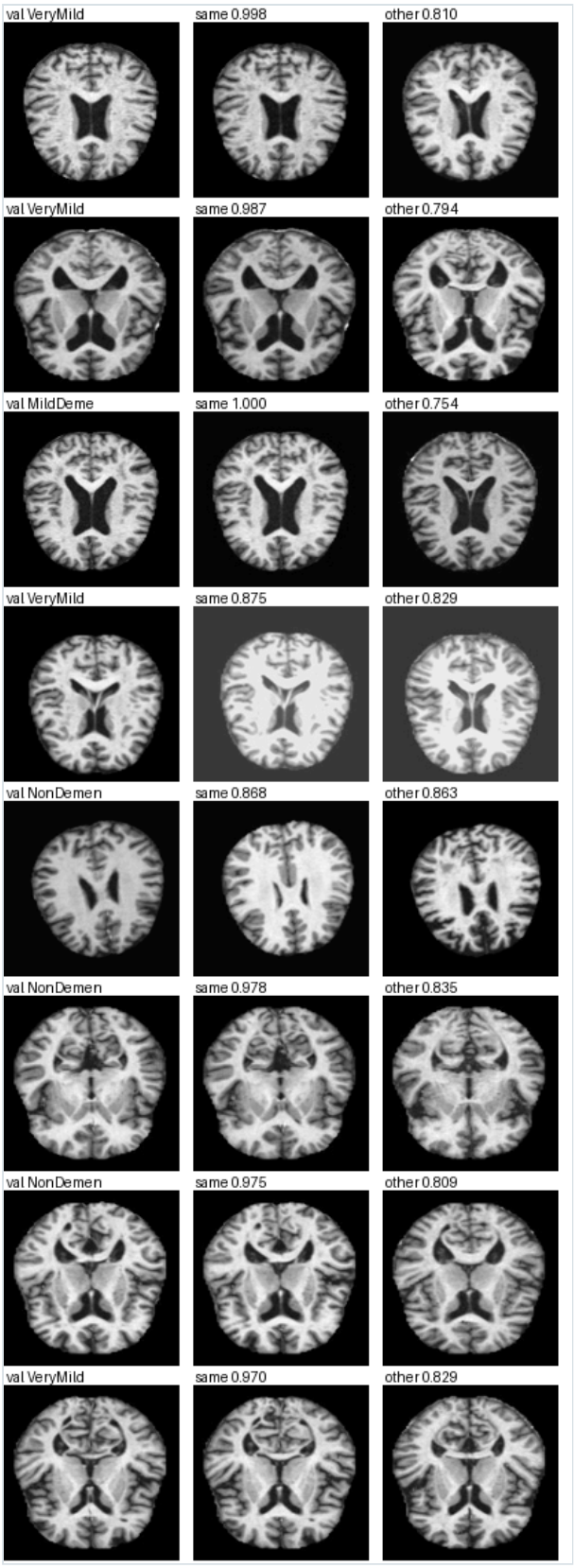
A faithful PyTorch port of `alzheimer_cnn.h5` was checked against Keras loading the same file. It needed three fixes the notebook missed: kernel layout, the order in which Keras flattens features, and the real 25088→256→4 head.

TRANSFER	PREDICTION AGREEMENT WITH KERAS	MAX PROBABILITY DIFFERENCE
Faithful port (`research/experiments/vgg_port.py`)	100%	0.0018 (Keras computes in float16)
Notebook cell 18 (reproduced)	17%	0.75

The notebook's transfer copied the 13 conv kernels but no biases and no classifier weights, so the original SNN ran on a mostly random network. A side finding: the saved model was compiled with a different loss than the notebook's training cell shows, so the `.h5` came from a different version of the code.

2. Leakage in the original dataset (E02–E02c, E08)

The first two attempts to detect near-copies automatically failed. Pooled VGG16 features made every slice look alike, and my calibration augmentations didn't match the dataset's. Visual inspection settled it. In each row below, the left image is from the validation set, the middle is its nearest same-class training image, and the right is the nearest image from a different class.



Eight random validation images. In six of them, the nearest same-class training image (similarity 0.970–1.000) is visibly the same slice. Every different-class neighbour is a different brain.

PIXEL SIMILARITY THRESHOLD	VALIDATION IMAGES WITH A SAME-CLASS TRAINING IMAGE ABOVE IT	DIFFERENT-CLASS NEIGHBOURS ABOVE IT (CONTROL)
0.93 (visually confirmed)	67.3%	0.0%
0.97	55.4%	0.0%
0.99	37.8%	0.0%

Effect on accuracy (E08). The same classifier, on frozen VGG16 features with 5 folds, was scored on two kinds of split:

SPLIT	MEAN ACCURACY
Random image split (as in the notebook)	79.3%
Near-copies kept together (threshold 0.97)	77.1%
Near-copies kept together (threshold 0.93)	73.6%

This is a **lower bound**. Flipped or rotated copies, and different slices from the same patient, can't be linked because the dataset has no patient IDs. This leakage is already documented in the literature (Yagis et al., 2021), so it is a correction to the paper, not a contribution.

3. Statistics for the genetics question (E03, E07)

E03 simulated 500 cohorts (n = 400) per scenario and tested whether APOE and a polygenic risk score add value to a learned MRI model. The table shows how often the test rejected at 5%:

SCENARIO	IN-SAMPLE MRI SCORE	OUT-OF-FOLD MRI SCORE	TRUE ATROPHY
No genetic effect at all (should be 5%)	4.0%	5.6%	4.0%
APOE acts only through atrophy	5.6%	85.6%	6.8%
Direct genetic effect (power)	32.0%	58.6%	69.8%

Implications. Out-of-fold MRI scores are mandatory: in-sample scoring loses about half the power. A significant result means "genetics adds information this MRI model missed", not "genetics acts directly", and

the paper must state this estimand in advance. It may also explain why published studies disagree: added value depends on how good each study's MRI model is.

E07 and E07b tested double machine learning, a method borrowed from econometrics, as a way to isolate a direct genetic effect. The core code is correct: it gives 6.0% false positives when handed the true helper functions. With realistic models, it still fails:

N	NO EFFECT	MEDIATED ONLY (SHOULD BE 5%)	DIRECT EFFECT (POWER)
400	4.3%	44%	60.3%
1,600	4.0%	28%	100%

Hypothesis falsified at realistic sample sizes. The method is demoted to a possible sensitivity analysis.

4. Testing saliency against anatomy (E05)

This used 200 synthetic smooth saliency maps that are unrelated to a hippocampus-shaped mask by construction, so a valid test should reject 5% of the time.

CONDITION	SHUFFLE VOXELS	SHIFT THE MASK
No relationship (should be 5%)	34.5%	5.0%
Weak true signal in the mask (power)	61% (not trustworthy)	15%

Shuffling voxels makes chance overlap look significant about a third of the time. The valid mask-shift test has low power for weak signals, so real analyses need strong effects or many subjects.

5. M5 hardware (E06)

WORKLOAD	CPU	GPU (MPS)	GPU + FP16
3D CNN training, 96×112×96 volumes	1.8 volumes/s	3.5 volumes/s	3.3 volumes/s
VGG16 2D inference	31 images/s	126 images/s	n/a

The GPU is compute-bound: throughput stays flat from batch 2 to batch 8, and memory isn't the limit (5.6 of 24 GB). An ADNI-scale run of 5 folds × 50 epochs would take about 12 hours. Because cost grows with voxel count, compact 3D representations are worth testing. No energy figures were measured or claimed.

Novelty ledger

ITEM	STATUS
Kaggle dataset leakage	Known; reproduced
Out-of-fold scoring for the added-value test	Known in principle; supported here
The added-value test can't separate mediated from direct genetic effects	Supported (synthetic); framing not found in 2 searches
Double machine learning for a direct genetic effect	Not supported at realistic n
Spatially aware nulls for saliency overlap	Known for brain maps; **possibly new** for AI saliency maps (needs a targeted search)
Attenuation curve (genetics' value vs MRI model capacity)	**Unverified**; now the lead candidate

Revised direction

- **Lead:** the attenuation curve. Measure genetics' added value against MRI models of increasing capacity (hippocampal volume → regional volumes → full 3D model). If APOE acts mostly through atrophy, its added value should shrink as the MRI model improves. This needs no causal claims, and it gives a concrete explanation for the conflicting published results. Next experiment: E09, in simulation.
- **Supporting:** a valid saliency-anatomy overlap test, and a comparison of compact and full 3D representations on the M5.
- **Dropped from the paper's main line:** the double machine learning direct-effect test, the SNN without hardware, and every result from the Kaggle dataset as evidence of generalisation.

Failures and corrections

- E02's embedding couldn't tell images apart, so its 99.9% near-copy rate is meaningless and must not be quoted. It was replaced by E02c's visually checked analysis.
- E02b's calibration used flips and rotations that don't match this dataset. Diagnosed and replaced.
- E07b's first run crashed on a bug I introduced while refactoring. It was fixed and rerun; no reported number comes from the broken run.
- E06 falsified my expectations of a large GPU gain and an fp16 speedup.
- The notebook itself was left unedited to preserve the original. The exact 86.16% can't be re-measured because the original split is unrecoverable.

What's needed to continue

1. Apply for **ADNI** access, including genetic data. This is the main blocker.
2. Apply for **AIBL or OASIS-3** as the external cohort.
3. Decide whether OASIS-1 (public 3D MRI under a data-use agreement) may be downloaded for developing the 3D pipeline.
4. Choose GWAS summary statistics for the polygenic risk score before any real results are seen.

Files

Everything is in `~/Desktop/CV EL/research/`: `ROADMAP.md`, `EXPERIMENTS.md`, `NOVELTY_LEDGER.md`, `HARDWARE.md` and `PUBLICATION_STATUS.md`. Experiment scripts are in `experiments/` and JSON records in `results/`. The tested pipeline code (cohort interface, calibration, saliency overlap, double machine learning) is in `pipeline/`. Rerun any experiment with `novelty/.venv/bin/python -I research/experiments/<script>.py`.

Prepared with Claude Code. Results from synthetic data check methods only and are not clinical evidence.